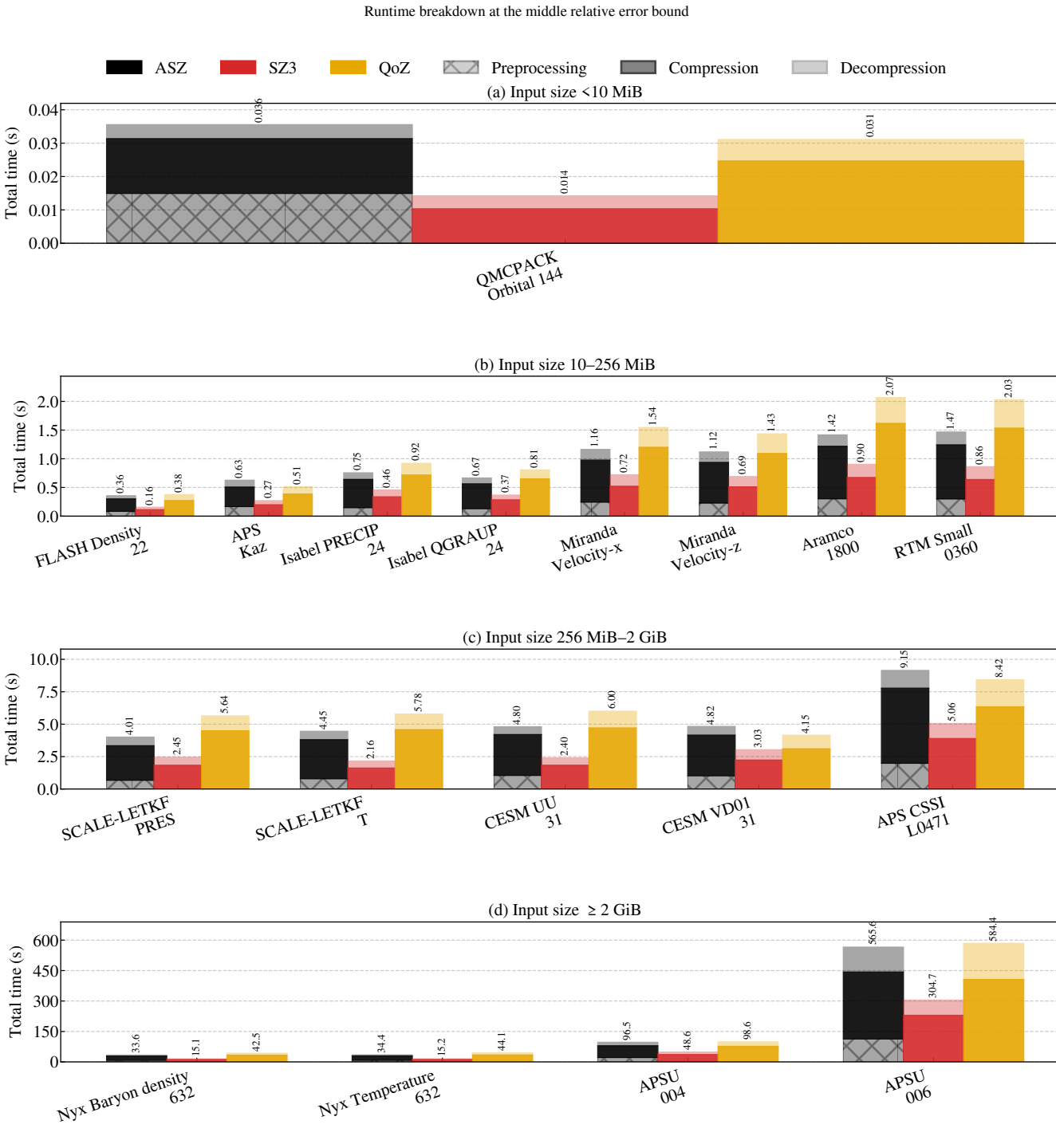
Runtime breakdown at the middle relative error bound
ASZ
SZ3
QoZ
Preprocessing
Compression
Decompression
(a) Input size <10 MiB
Total time (s)
0.036
0.014
0.031
QMCPACK Orbital 144
(b) Input size 10–256 MiB
0.36
0.16
0.38
0.63
0.27
0.51
0.75
0.46
0.92
0.67
0.37
0.81
1.16
0.72
1.54
1.12
0.69
1.43
1.42
0.90
2.07
1.47
0.86
2.03
FLASH Density 22
APS Kaz
Isabel PRECIP 24
Isabel QGRAUP 24
Miranda Velocity-x
Miranda Velocity-z
Aramco 1800
RTM Small 0360
(c) Input size 256 MiB–2 GiB
4.01
2.45
5.64
4.45
2.16
5.78
4.80
2.40
6.00
4.82
3.03
4.15
9.15
5.06
8.42
SCALE-LETKF PRES
SCALE-LETKF T
CESM UU 31
CESM VD01 31
APS CSSI L0471
(d) Input size ≥ 2 GiB
33.6
15.1
42.5
34.4
15.2
44.1
96.5
48.6
98.6
565.6
304.7
584.4
Nyx Baryon density 632
Nyx Temperature 632
APSU 004
APSU 006